MTH 207 Practice Exam 2

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Rules

- The cover page should only have your name and email. Nothing else may may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating "does not exist" without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

1. Find $\lim_{x\to 0} \frac{\sin(3x)\cdot 3}{4x}$

$= \lim_{x\to 0} \frac{3}{4}\cdot\frac{\sin(3x)}{3x}$

$= \frac{3}{4}\lim_{x\to 0}\frac{\sin(3x)}{3x}$ $\quad \theta = 3x \quad$ As $x\to 0 \quad \theta\to 0$

$= \frac{3}{4}\lim_{\theta\to 0}\frac{\sin(\theta)}{\theta}$

$= \frac{3}{4}\cdot 1$

$= \frac{3}{4}$

2. Find $\lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h}\cdot\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}$

$= \lim_{h\to 0}\frac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})}$

$= \lim_{h\to 0}\frac{h}{h(\sqrt{x+h}+\sqrt{x})}$

$= \lim_{h\to 0}\frac{1}{\sqrt{x+h}+\sqrt{x}}$

$= \frac{1}{\sqrt{x+0}+\sqrt{x}} = \frac{1}{2\sqrt{x}}$

3. Find $\lim_{x\to 3} \frac{x-3}{x^2-9}$

Can't split

$\lim_{x\to 3}(x-3) = 3-3 = 0 \quad \lim_{x\to 3}(x^2-9) = 3^2-9 = 0$

$= \lim_{x\to 3}\frac{(x-3)}{(x+3)(x-3)}$

$= \lim_{x\to 3}\frac{1}{x+3}$ $\quad \lim_{x\to 3} 1 = 1 \quad \lim_{x\to 3}(x+3) = 3+3 = 6$

Can split

$= \frac{\lim_{x\to 3} 1}{\lim_{x\to 3}(x+3)}$

$= \frac{1}{6}$

Same function as Q3 but derivative

4. If $f(x) = \dfrac{x-3}{x^2-9}$, find $f'(x)$

$$f'(x) = \frac{(x^2-9)(1-0) - (x-3)(2x-0)}{(x^2-9)^2}$$

5. If $y = \sqrt{x^9 - 10x^7 + 5^x}$ find $\dfrac{dy}{dx}$

$$y = (x^9 - 10x^7 + 5^x)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2}(x^9 - 10x^7 + 5^x)^{-1/2} \cdot (9x^8 - 10 \cdot 7x^6 + 5^x \ln 5)$$

$\sin\left((\sin(x))^{\frac{1}{3}}\right)$

6. If $f(x) = x^3 \cdot 6^x + 5\cos(x) + \sin\left(\sqrt[3]{\sin(x)}\right)$, find $f'(x)$

$$f'(x) = x^3 \cdot 6^x \ln 6 + 6^x \cdot 3x^2 + 5(-\sin x) + \cos\left((\sin x)^{1/3}\right) \cdot \frac{1}{3}(\sin(x))^{-2/3} \cdot \cos(x)$$

7. If $y = 3x^5 \cdot 7^{\sin(x)+\cot(x)}$ find $\frac{dy}{dx}$

$$\frac{dy}{dx} = 3x^5 \cdot 7^{\sin(x)+\cot(x)} \ln(7) \cdot (\cos x + (-\csc^2 x)) + 7^{\sin x + \cot x} \cdot (3 \cdot 5x^4)$$

8. If $f(x) = 4\tan(x) + 6^x + \frac{x^3+x^2+x+1}{x}$, find $f''(x)$

Algebra first

$$f(x) = 4\tan x + 6^x + x^2 + x + 1 + x^{-1}$$

$$f'(x) = 4\sec^2 x + 6^x \ln 6 + 2x + 1 + 0 + (-1)x^{-2}$$

Copy down to Constant Mult. Rule

Copy (same name)

$$f''(x) = 4 \cdot 2(\sec x)(\sec x \tan x) + (6^x \ln 6) \cdot \ln 6 + 2 \cdot 1 + 0 + 0 + (-1) \cdot (-2)x^{-3}$$

9. If $y = \frac{3 + x^9 \sin(4x)}{2^x + x^9} + \sin\left(\frac{x^2}{x^4+1}\right)$ find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{(2^x + x^9)(0 + x^9 \cos(4x) \cdot x + \sin(4x) \cdot 9x^8) - (3 + x^9 \sin(4x))(2^x \ln 2 + 9x^8)}{(2^x + x^9)^2} + \cos\left(\frac{x^2}{x^4+1}\right) \cdot \frac{(x^4+1)(2x) - (x^2)(4x^3+0)}{(x^4+1)^2}$$

10. If $f(x) = 0.1x^2 + x$ describes the position (in meters) of Brewers coach Pat Murphy's broom x minutes after 12 : 00 noon, what is the velocity of Murphy's sweeping 10 minutes after noon?

$f'(x) = 0.1 \cdot 2x + 1$

$f'(10) = 0.1 \cdot 2(10) + 1$ no requirement to simplify, but if you do...

$= 0.1(20) + 1 = 2 + 1 = 3$ meters per minute.

11. If $f(x) = x^2 + \sin(x)$, find $f'(\frac{4\pi}{3})$ and **SIMPLIFY**.

$f'(x) = 2x + \cos(x)$

$f'\left(\frac{4\pi}{3}\right) = 2 \cdot \frac{4\pi}{3} + \cos\left(\frac{4\pi}{3}\right)$

$= \frac{8\pi}{3} + -\frac{1}{2}$

$= \frac{16\pi - 3}{6}$

12. If y depends on x according ~~$\frac{d}{dx}$~~$\left(x^3 + y^5\right) = $ ~~d~~$\left(x + \sin(x^7 y^9)\right)$, find $\frac{dy}{dx}$.

$3x^2 + 5y^4 \cdot \frac{dy}{dx} = 1 + \cos(x^7 y^9) \cdot (x^9 \cdot 9y^8 \cdot \frac{dy}{dx} + y^9 \cdot 7x^6)$

$3x^2 + 5y^4 \cdot \frac{dy}{dx} = 1 + 9x^9 y^8 \cos(x^7 y^9) \cdot \frac{dy}{dx} + 7x^6 y^9 \cos(x^7 y^9)$

$5y^4 \cdot \frac{dy}{dx} - 9x^9 y^8 \cos(x^7 y^9) \cdot \frac{dy}{dx} = 1 + 7x^6 y^9 \cos(x^7 y^9) - 3x^2$

$\frac{dy}{dx}\left(5y^4 \cdot - 9x^9 y^8 \cos(x^7 y^9)\right) = 7x^6 y^9 \cos(x^7 y^9) - 3x^2$

$\frac{dy}{dx} = \frac{7x^6 y^9 \cos(x^7 y^9) - 3x^2}{5y^4 \cdot - 9x^9 y^8 \cos(x^7 y^9)}$

13. Find an equation for the tangent line to $f(x) = 2x^4$ at $x = 10$.

$f'(x) = 2 \cdot 4x^3 = 8x^3$

$f'(10) = 8 \cdot (10)^3 = 8000$

$f(10) = 2(10)^4 = 20000$

$$y - 20000 = 8000(x - 10)$$

14. Starting from $y = \sec(x)$ find a formula for $\frac{dy}{dx}$, showing all work including a relevant trig identity.

$y = \frac{1}{\cos x}$

$\frac{dy}{dx} = \frac{(\cos x)(0) - (1)(-\sin x)}{(\cos x)^2} = \frac{\sin x}{\cos x \cdot \cos x} = \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x}$

$= \tan x \cdot \sec x$

15. Starting from $y = \arccos(x)$ find a formula for $\frac{dy}{dx}$, showing all work including a relevant trig identity.

$x = \cos(y)$

$\frac{d}{dx} \downarrow$

$1 = -\sin(y) \cdot \frac{dy}{dx}$

$\frac{dy}{dx} = \frac{1}{-\sin y}$

$\sin^2 y + \cos^2 y = 1$

$\sin y = \sqrt{1 - \cos^2 y}$

$\frac{dy}{dx} = \frac{1}{-\sqrt{1 - \cos y}}$ ← still has y, but $\cos(y) = x$

$\frac{dy}{dx} = \frac{1}{-\sqrt{1 - x^2}}$